

Md. Basim Al Zabir Shammo

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RESEARCH INTERESTS

Physics-Informed Machine Learning for Digital Twins, Multi-Modal Sensor Fusion Prognostics, and Vision-Based Autonomous Robotics

EDUCATION

Pabna University of Science and Technology, B.Sc. (Engineering) in Information and Communication Engineering Feb 2022 – Sep 2026

- CGPA: 3.36/4.0

EXPERIENCE

Research Assistant, Pabna University of Science and Technology Jan 2025 – Jul 2026

- **Mathematical Modeling:** Modeled dynamic loads via electrical impedance (R_{eq}^{mech}), eliminating physical encoders in edge-deployed digital twins.
- **Sensor Fusion & Health Index:** Engineered a Thermal Health Index (THI) by fusing multi-modal telemetry and analyzing GRU prediction residuals.
- **Prognostics & Uncertainty:** Forecasted thermal saturation thresholds under varying loads, quantifying epistemic uncertainty via Monte Carlo Dropout.
- **Applied Machine Learning:** Co-authored 6 papers on ML-based diagnostic models using PPG signal analysis and ensemble boosting.
- **Robotics & Simulation:** Integrated YOLOv11n-seg with zig-zag coverage; built an IR-vision fusion algorithm resolving deadlocks to achieve 94.17% Webots simulation coverage.
- **Grant Writing:** Drafted robotics grant proposals and defined technical specifications for AI lab equipment.

PUBLICATIONS

Manuscripts in Revision & Under Review

- **Md. Basim Al Zabir Shammo**, Md. Anwar Hossain, et al. "The Limits of Physics-Informed Digital Twins Under Hardware-Constrained Sampling: An HIL Study for Micro-Mechatronics" Journal of Intelligent Manufacturing (SCIE Indexed, JCR: Q1, Scopus: Q1). Springer Nature. (3rd Revision Submitted)
- **Md. Basim Al Zabir Shammo**, Tarun Debnath, et al. "Vision-Based Crack Detection and Coverage Planning for Autonomous Mobile Robots." International Journal of Intelligent Robotics and Applications (Scopus: Q2). Springer Nature. (1st Revision Ongoing)
- Noor Islam S. Mohammad, **Md. Basim Al Zabir Shammo**, et al. "EthosVLA: Constitutional Safety for Embodied Vision-Language Agents." 2027 International Conference on Learning Representations (ICLR). (Submitted)
- Noor Islam S. Mohammad, **Md. Basim Al Zabir Shammo**, et al. "SIFT: Robust Meta-Faithfulness Verification of Chain-of-Thought Reasoning Under Distribution Shift." 2027 International Conference on Learning Representations (ICLR). (Submitted)
- Noor Islam S. Mohammad, . . . , **Md. Basim Al Zabir Shammo**, et al. "The Symmetry Budget: How Much Group Structure Should a Network Architecture Assume?" 2026 Conference on Neural Information Processing Systems (NeurIPS) Workshop. (Submitted)
- Noor Islam S. Mohammad, . . . , **Md. Basim Al Zabir Shammo**. "Neural Representations for Universal Speech Restoration via Corruption-Conditioned Diffusion Models." 2026 Conference on Neural Information Processing Systems (NeurIPS) Workshop. (Submitted)

Published Journals

- Jakaria Habib, Md. Imran Hossain, **Md. Basim Al Zabir Shammo**, et al. "Vision Transformer-based Damage Assessment from Post-Disaster Satellite Imagery: An Applied Study on Hurricane Harvey." Earth Science Informatics (SCIE Indexed, Scopus: Q1). Springer Nature. [DOI](#)

Published Conferences

- **Md. Basim Al Zabir Shammo**, Lamia Islam, et al. “**Machine Learning-Based Hepatitis C Detection Using Ensemble Boosting.**” 2025 International Conference on Big Data, IoT and Machine Learning (BIM), Dhaka, Bangladesh, Lecture Notes in Networks and Systems (Scopus: Q4). Springer Nature [DOI](#)
- Lamia Islam, **Md. Basim Al Zabir Shammo**, et al. “**Machine Learning-Based Prediction of Nightmares in Children: A Pediatric Psychological Integration.**” 2025 IEEE Conference on Computing Applications and Systems (COMPAS), Kushtia, Bangladesh. IEEE Xplore. [DOI](#)
- Md. Rashed, **Md. Basim Al Zabir Shammo**, et al. “**A Machine Learning Framework for Stress Detection Using Photoplethysmography (PPG) Signals.**” 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh. IEEE Xplore. [DOI](#)
- Lamia Islam, **Md. Basim Al Zabir Shammo**, et al. “**Real-Time RFID Authentication and Access Control System with Networked Logging.**” 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN), Chattogram, Bangladesh. [DOI](#)
- Md. Akash Miya, . . . , **Md. Basim Al Zabir Shammo**, et al. “**Remote Health Monitoring via PPG Signal Processing and Machine Learning using Mobile Devices.**” 28th International Conference on Computer and Information Technology (ICCIT), 2025. IEEE Xplore. [DOI](#)
- Naima Islam, . . . , **Md. Basim Al Zabir Shammo**. “**Vehicle Classification and Detection Using YOLOv8: A Study on Highway Traffic Analysis.**” 2024 International Conference on Recent Progresses in Science, Engineering and Technology (ICRPSET), Rajshahi, Bangladesh. IEEE Xplore. [DOI](#)

ACADEMIC SERVICE

- [Peer Reviewer](#), 28th International Conference on Computer and Information Technology (ICCIT) 2025
- [Peer Reviewer](#), 4th IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON) 2025

TECHNICAL SKILLS

- **Embedded Systems & Hardware Integration:** Hardware-in-the-Loop (HIL) Simulation, Multi-Sensor Fusion, Microcontrollers (ESP32, ESP8266, Arduino), 3D Printing
- **Robotics & Computer Vision:** Mobile Robotics Simulation in Webots, IR-Vision Fusion, Object Detection & Instance Segmentation using YOLOv8, YOLOv11n-seg and OpenCV.
- **Artificial Intelligence & Machine Learning:** Applied Machine Learning, Predictive Modeling, Ensemble Boosting, Time-Series Analysis using GRU, Uncertainty Quantification using Monte Carlo Dropout, Scikit-learn, TensorFlow, PyTorch, Keras
- **Programming & Modeling:** Python, R, C, Mathematical Modeling of Dynamic Loads, Prognostics & Health Management

HONORS

- Recipient, PUST Conference and Travel Grant
- Finalist, 2025 World Robotics Olympiad Bangladesh
- Recipient, 2022 Earth Champions Program (ECP) Fellowship, Bangladesh Youth Environmental Initiative (BYEI), Fully Funded (100,000 BDT)

REFERENCES

Dr. Md. Mahbubur Rahman
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Military Institute of Science and Technology
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Dr. Md. Anwar Hossain
Professor, Dept. of ICE
Pabna University of Science and Technology
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